

Supporting Information

Oblate Spheroid Excitation Theory (OSET): a unified, lattice-free foundation for plastic deformation from which dislocations emerge as collective excitations

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This document gives the complete step-by-step standard (non-novel) derivations and the mechanical algebra on which the main-text results rest. Equation, figure and table numbers are prefixed “S”; references to main-text equations are written “Eq. (main)”.

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S1 Eshelby tensor for the oblate spheroid

By Eshelby's theorem the strain inside a uniformly eigenstrained ellipsoidal inclusion in an isotropic matrix is uniform, $\varepsilon_{ij}^{\text{in}} = S_{ijkl}\varepsilon_{kl}^*$, with the Eshelby tensor expressible through the elliptic I -integrals [1]

$$I_\alpha = 2\pi a_1 a_2 a_3 \int_0^\infty \frac{ds}{(a_\alpha^2 + s)\Delta(s)}, \quad (S1)$$

$$I_{\alpha\beta} = 2\pi a_1 a_2 a_3 \int_0^\infty \frac{ds}{(a_\alpha^2 + s)(a_\beta^2 + s)\Delta(s)}, \quad \Delta(s) = \sqrt{(a_1^2 + s)(a_2^2 + s)(a_3^2 + s)}, \quad (S2)$$

which satisfy the sum rule $I_1 + I_2 + I_3 = 4\pi$. The shear component required for the OSTZ is

$$S_{1313} = \frac{(1 + \alpha^2) I_{13} + (1 - 2\nu)(I_1 + I_3)}{16\pi(1 - \nu)}. \quad (S3)$$

The stress inside the inclusion is uniform because the Newtonian potential of a uniform-density ellipsoid is a quadratic polynomial inside the body; the elastic strain, set by its second derivatives, is therefore constant. Ellipsoids are the only shapes for which S_{ijkl} is spatially uniform.

S1.1 Closed-form evaluation of I_1

Specialise Eq. (S1) to the oblate spheroid $a_1 = a_2 = 1$, $a_3 = \alpha < 1$, so $\Delta(s) = (1 + s)\sqrt{\alpha^2 + s}$ and

$$I_1 = 2\pi\alpha \int_0^\infty \frac{ds}{(1 + s)^2 \sqrt{\alpha^2 + s}}. \quad (S4)$$

Substitution. Let $t = \sqrt{\alpha^2 + s}$, so $s = t^2 - \alpha^2$, $ds = 2t dt$; the lower limit $s = 0$ maps to $t = \alpha$ and $s \rightarrow \infty$ to $t \rightarrow \infty$. Writing $e^2 \equiv 1 - \alpha^2$ gives $1 + s = t^2 + e^2$, hence

$$I_1 = 2\pi\alpha \int_\alpha^\infty \frac{2t dt}{(t^2 + e^2)^2 t} = 4\pi\alpha \int_\alpha^\infty \frac{dt}{(t^2 + e^2)^2}. \quad (S5)$$

Antiderivative. The standard reduction (verified below) is

$$\int \frac{dt}{(t^2 + e^2)^2} = \frac{t}{2e^2(t^2 + e^2)} + \frac{1}{2e^3} \arctan \frac{t}{e} + C. \quad (S6)$$

Proof. With $F = t/[2e^2(t^2 + e^2)]$ and $G = \arctan(t/e)/(2e^3)$,

$$F' = \frac{e^2 - t^2}{2e^2(t^2 + e^2)^2}, \quad G' = \frac{1}{2e^2(t^2 + e^2)}, \quad F' + G' = \frac{(e^2 - t^2) + (t^2 + e^2)}{2e^2(t^2 + e^2)^2} = \frac{1}{(t^2 + e^2)^2}. \quad (S7)$$

Bounds. At $t \rightarrow \infty$ the first term vanishes and $\arctan(t/e) \rightarrow \pi/2$, giving $\pi/(4e^3)$. At $t = \alpha$, using $\alpha^2 + e^2 = 1$, the bracket is $\alpha/(2e^2) + \arctan(\alpha/e)/(2e^3)$. Subtracting,

$$\int_\alpha^\infty \frac{dt}{(t^2 + e^2)^2} = \frac{1}{2e^3} \left[\frac{\pi}{2} - \arctan \frac{\alpha}{e} \right] - \frac{\alpha}{2e^2}, \quad (S8)$$

so that

$$I_1 = \frac{2\pi\alpha}{e^3} \left[\frac{\pi}{2} - \arctan \frac{\alpha}{e} - e\alpha \right]. \quad (S9)$$

Trigonometric identity. Setting $\alpha = \cos \phi$ ($e = \sin \phi$, $\phi = \arccos \alpha$) gives $\arctan(\alpha/e) = \arctan(\cot \phi) = \pi/2 - \phi$, hence $\frac{\pi}{2} - \arctan(\alpha/e) = \arccos \alpha$. With $e^3 = (1 - \alpha^2)^{3/2}$,

$$I_1 = \frac{2\pi\alpha}{(1 - \alpha^2)^{3/2}} \left[\arccos \alpha - \alpha\sqrt{1 - \alpha^2} \right], \quad I_3 = 4\pi - 2I_1. \quad (\text{S10})$$

S1.2 The relation $I_{13} = (I_3 - I_1)/(1 - \alpha^2)$

With $a_1 = a_2 = 1$, $a_3 = \alpha$,

$$I_3 - I_1 = 2\pi\alpha \int_0^\infty \frac{1}{(1+s)\sqrt{\alpha^2+s}} \left[\frac{1}{\alpha^2+s} - \frac{1}{1+s} \right] ds. \quad (\text{S11})$$

The bracket equals $\frac{(1+s) - (\alpha^2+s)}{(\alpha^2+s)(1+s)} = \frac{1-\alpha^2}{(\alpha^2+s)(1+s)}$, so

$$I_3 - I_1 = (1 - \alpha^2) 2\pi\alpha \int_0^\infty \frac{ds}{(1+s)^2(\alpha^2+s)^{3/2}} = (1 - \alpha^2) I_{13} \Rightarrow I_{13} = \frac{I_3 - I_1}{1 - \alpha^2}. \quad (\text{S12})$$

S1.3 Numerical evaluation at $\alpha = 1/2$, $\nu = 1/3$

With $e = \sqrt{0.75} = 0.8660$,

$$I_1 = \frac{2\pi(0.5)}{0.75^{3/2}} [\arccos 0.5 - 0.5(0.8660)] = 4.835 \times 0.6142 = 2.972, \quad (\text{S13})$$

$$I_3 = 4\pi - 2(2.972) = 6.622, \quad I_{13} = \frac{6.622 - 2.972}{0.75} = 4.867, \quad (\text{S14})$$

$$S_{1313} = \frac{1.25(4.867) + \frac{1}{3}(9.594)}{16\pi(\frac{2}{3})} = \frac{6.084 + 3.198}{33.51} = 0.2772, \quad \beta_1 = 1 - 2(0.2772) = 0.4456. \quad (\text{S15})$$

S1.4 Spherical-mode contraction and β_2

For the dilatational OSTZ mode ($\varepsilon_{ij}^* = (\varepsilon_0/3)\delta_{ij}$) the sphere Eshelby tensor obeys $S_{1111} = \frac{7-5\nu}{15(1-\nu)}$, $S_{1122} = \frac{5\nu-1}{15(1-\nu)}$, so

$$S_{ijkk} = (S_{1111} + 2S_{1122})\delta_{ij} = \frac{(7-5\nu) + 2(5\nu-1)}{15(1-\nu)}\delta_{ij} = \frac{1+\nu}{3(1-\nu)}\delta_{ij}. \quad (\text{S16})$$

The interior dilatation is then $\varepsilon_{ij}^{\text{in}} = \frac{\varepsilon_0}{3}S_{ijkk} = \frac{(1+\nu)\varepsilon_0}{9(1-\nu)}\delta_{ij}$, and the elastic part

$$e_{ij}^{\text{el}} = \varepsilon_{ij}^{\text{in}} - \frac{\varepsilon_0}{3}\delta_{ij} = \varepsilon_0 \left[\frac{1+\nu}{9(1-\nu)} - \frac{1}{3} \right] \delta_{ij} = -\frac{2(1-2\nu)\varepsilon_0}{9(1-\nu)}\delta_{ij}, \quad (\text{S17})$$

with trace $e_{kk}^{\text{el}} = -\frac{2(1-2\nu)}{3(1-\nu)}\varepsilon_0$. Carrying this through the inclusion-energy theorem (Eq. main) gives the dilatational factor

$$\beta_2 = \frac{4(1+\nu)}{9(1-\nu)} \Rightarrow \beta_2(\nu = \frac{1}{3}) = 0.889. \quad (\text{S18})$$

S1.5 Limiting cases

Sphere ($\alpha \rightarrow 1$). $I_1 = I_2 = I_3 = 4\pi/3$ by symmetry and the sum rule; with $\Delta = (1+s)^{3/2}$, $I_{13} = 2\pi \int_1^\infty u^{-7/2} du = 4\pi/5$. Then

$$S_{1313}^{\text{sphere}} = \frac{2(\frac{4\pi}{5}) + (1-2\nu)\frac{8\pi}{3}}{16\pi(1-\nu)} = \frac{1}{2(1-\nu)} \left[\frac{1}{5} + \frac{1-2\nu}{3} \right] = \frac{4-5\nu}{15(1-\nu)}, \quad (\text{S19})$$

i.e. 0.2333 at $\nu = 1/3$, with $S_{1313} = S_{1212}$ as required by isotropy.

Thin disc ($\alpha \rightarrow 0$). $I_1 \rightarrow 0$, $I_3 \rightarrow 4\pi$, $I_{13} \rightarrow 4\pi$, giving $S_{1313}^{\text{disc}} = \frac{4\pi+(1-2\nu)4\pi}{16\pi(1-\nu)} = \frac{1}{2}$, $\beta_1 \rightarrow 0$: a crack-like disc stores no shear energy.

Table S1: Numerically verified Eshelby components and shear constraint factor for the oblate spheroid $\alpha = 1/2$.

ν	S_{1313}	S_{1212}	S_{3333}	$\beta_1 = 1 - 2S_{1313}$
0.30	0.2822	0.1769	0.7264	0.4356
1/3	0.2772	0.1739	0.7364	0.4456
0.35	0.2745	0.1723	0.7418	0.4510
0.40	0.2656	0.1670	0.7596	0.4688

S1.6 The two constraint regimes of β_1

The Eshelby value $\beta_1^{\text{Esh}} = 1 - 2S_{1313} \approx 0.44\text{--}0.49$ ($\nu = 0.28\text{--}0.44$) applies to a zone in homogeneous bulk. OSTZs in grain-boundary material of reduced modulus $G_{\text{GB}} \approx G/2$ feel a larger effective constraint, $\beta_1^{\text{eff}} \approx \beta_1^{\text{Esh}} G/G_{\text{GB}} \approx 0.9 \approx 1$, reconciling the elastic value with the $\beta_1^{\text{eff}} = 1$ used for coherent faults; single-crystal interiors use $\beta_1^{\text{Esh}} \approx 0.45$.

S2 Far-field dipole: complete Kelvin Green's-function derivation

S2.1 Why the monopole vanishes and the dipole survives

The exterior field of a finite inclusion has the multipole structure $\sigma_{ij} = F_k \mathcal{G}_{ijk} + M_{kl} \partial_l \mathcal{G}_{ijk} + \dots$. The net force is $F_k = \oint_{\partial V_0} \sigma_{ij}^* n_j dS = C_{ijkl} \varepsilon_{kl}^* \oint_{\partial V_0} n_j dS = 0$, because $\oint n_j dS = \int_{V_0} \partial_j(1) dV = 0$, so there is no r^{-2} monopole. The dipole (moment) tensor is $M_{kl} = C_{ijkl} \varepsilon_{ij}^* V_0$; for the OSTZ eigenstrain $\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2$ this gives $M_{13} = M_{31} = G\gamma_0 V_0$, a double couple producing the leading r^{-3} field.

S2.2 Step A: the Kelvin Green's function and its derivative

The displacement at $\mathbf{r}$ from a unit point force at the origin is

$$\mathcal{G}_{ij} = A \left[\frac{(3-4\nu)\delta_{ij}}{r} + \frac{x_i x_j}{r^3} \right], \quad A = \frac{1}{16\pi G(1-\nu)}. \quad (\text{S20})$$

Using $\partial_l r = x_l/r$, $\partial_l(1/r) = -x_l/r^3$, $\partial_l x_i = \delta_{il}$ and $\partial_l(r^{-3}) = -3x_l/r^5$, differentiation of the two terms gives

$$\partial_l \mathcal{G}_{ij} = A \left[-\frac{(3-4\nu)\delta_{ij} x_l}{r^3} + \frac{\delta_{il} x_j + \delta_{jl} x_i}{r^3} - \frac{3x_l x_i x_j}{r^5} \right]. \quad (\text{S21})$$

S2.3 Step B: the displacement field

The exterior displacement from the moment tensor is $u_i = -M_{kl} \partial_l \mathcal{G}_{ik}$ (the sign is the Eshelby exterior convention). With $M_{13} = M_{31} = G\gamma_0 V_0$ only $(k, l) = (1, 3), (3, 1)$ contribute,

$$u_i = -G\gamma_0 V_0 [\partial_3 \mathcal{G}_{i1} + \partial_1 \mathcal{G}_{i3}]. \quad (\text{S22})$$

Adding the two copies of Eq. (S21), the $\delta_{i1}x_3$ terms combine with coefficient $1 - (3 - 4\nu) = 2(2\nu - 1)$ (and likewise $\delta_{i3}x_1$), and the two r^{-5} pieces add to $-6x_1x_3x_i/r^5$, giving

$$u_i = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)(\delta_{i1}x_3 + \delta_{i3}x_1)}{r^3} - \frac{6x_1x_3x_i}{r^5} \right]. \quad (\text{S23})$$

The relevant components are

$$u_1 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)x_3}{r^3} - \frac{6x_1^2x_3}{r^5} \right], \quad u_3 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)x_1}{r^3} - \frac{6x_1x_3^2}{r^5} \right]. \quad (\text{S24})$$

S2.4 Step C: the stress from displacement gradients

For the shear component, $\sigma_{13} = G(\partial_1 u_3 + \partial_3 u_1)$. Using $\partial_3(x_3/r^3) = (r^2 - 3x_3^2)/r^5$ and $\partial_3(x_3/r^5) = (r^2 - 5x_3^2)/r^7$ (and the $1 \leftrightarrow 3$ images),

$$\partial_3 u_1 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)(r^2 - 3x_3^2)}{r^5} - \frac{6x_1^2(r^2 - 5x_3^2)}{r^7} \right], \quad (\text{S25})$$

$$\partial_1 u_3 = -G\gamma_0 V_0 A \left[\frac{2(2\nu - 1)(r^2 - 3x_1^2)}{r^5} - \frac{6x_3^2(r^2 - 5x_1^2)}{r^7} \right]. \quad (\text{S26})$$

Adding and converting to direction cosines $\hat{r}_i = x_i/r$, the r^{-5} group gives $2(2\nu - 1)(2 - 3\hat{r}_1^2 - 3\hat{r}_3^2)/r^3$ and the r^{-7} group $[-6(\hat{r}_1^2 + \hat{r}_3^2) + 60\hat{r}_1^2\hat{r}_3^2]/r^3$. Using $6(2\nu - 1) + 6 = 12\nu$,

$$\partial_3 u_1 + \partial_1 u_3 = -\frac{G\gamma_0 V_0 A}{r^3} [4(2\nu - 1) - 12\nu(\hat{r}_1^2 + \hat{r}_3^2) + 60\hat{r}_1^2\hat{r}_3^2]. \quad (\text{S27})$$

S2.5 Step D: the angular tensor

Multiplying by G and writing $A = [16\pi G(1 - \nu)]^{-1}$,

$$\sigma_{13} = \frac{G\gamma_0 V_0}{16\pi(1 - \nu)r^3} [4 - 8\nu + 12\nu(\hat{r}_1^2 + \hat{r}_3^2) - 60\hat{r}_1^2\hat{r}_3^2] = \frac{G\gamma_0 V_0}{2\pi(1 - \nu)} \frac{\mathcal{T}_{13}(\hat{\mathbf{r}})}{r^3}, \quad (\text{S28})$$

where, dividing the bracket by 8 (the prefactor ratio $2\pi/16\pi$),

$$\boxed{\mathcal{T}_{13}(\hat{\mathbf{r}}) = \frac{1}{2} - \nu + \frac{3\nu}{2}(\hat{r}_1^2 + \hat{r}_3^2) - \frac{15}{2}\hat{r}_1^2\hat{r}_3^2.} \quad (\text{S29})$$

The two derivatives of $\mathcal{G}$ each lower the power of r by one, turning the r^{-1} displacement into the r^{-3} dipole stress.

S2.6 Step E: verification in special directions

On the glide plane ($\hat{r}_3 = 0$), $\mathcal{T}_{13} = \frac{1}{2} - \nu + \frac{3\nu}{2}\hat{r}_1^2$, which rearranges to

$$\sigma_{13}(x_1, x_2, 0) = \frac{G\gamma_0 V_0}{4\pi(1 - \nu)r^5} [(1 + \nu)x_1^2 + (1 - 2\nu)x_2^2], \quad (\text{S30})$$

positive definite for $0 < \nu < \frac{1}{2}$. On axis ($\hat{r}_1 = 0, \hat{r}_3 = 1$), $\mathcal{T}_{13} = (1 + \nu)/2 > 0$, identical to the value along $\hat{x}_1$ by the axial symmetry of the spheroid. No real nodal surface exists for $0 < \nu < \frac{1}{2}$, so σ_{13} does not change sign in the half-space — the cooperative-nucleus character of the OSTZ. The leading far-field structure reduces to the dipole-dipole form $\propto (3\cos^2\theta - 1)$.

S3 Lorentzian density: compliance, Abel projection, Fourier transform

S3.1 Step A: Bilby–Cottrell–Swinden planar compliance

A shear traction σ_{13} on a planar slip distribution in an isotropic solid produces local slip $\delta u_1 = 2(1-\nu)\sigma_{13}/G$; the factor $2(1-\nu)/G$ is the BCS shear compliance, equivalently the Peierls–Nabarro relation $\sigma = G\rho/[2(1-\nu)]$ with $\rho = du/dx$ [2]. Because the OSTZ stress decays as r^{-3} the compliance integral converges absolutely without a short-distance cut-off.

S3.2 Step B: two-dimensional planar density

With the regularised OSTZ glide-plane traction $\sigma_{13} = Gb_{\text{eff}}W^2/[2\pi(1-\nu)(x^2 + y^2 + W^2)^{3/2}]$, the compliance gives, after cancellation of G , $2(1-\nu)$ and the factor 2,

$$\rho_{2D}(x, y) = \frac{2(1-\nu)}{G} \sigma_{13} = \frac{b_{\text{eff}} W^2}{\pi(x^2 + y^2 + W^2)^{3/2}}. \quad (\text{S31})$$

A normalisation check in polar coordinates, $\iint \rho_{2D} dx dy = \frac{b_{\text{eff}}W^2}{\pi} 2\pi \int_0^\infty \frac{r dr}{(r^2+W^2)^{3/2}} = \frac{b_{\text{eff}}W^2}{\pi} 2\pi \frac{1}{W} = 2b_{\text{eff}}W$, recovers the slip moment (displacement $\times$ length).

S3.3 Step C: Abel projection to one dimension

Integrating over y with $a^2 = x^2 + W^2$ and $y = a \tan \theta$,

$$\int_{-\infty}^{\infty} \frac{dy}{(a^2 + y^2)^{3/2}} = \frac{1}{a^2} \int_{-\pi/2}^{\pi/2} \cos \theta d\theta = \frac{2}{a^2}, \quad (\text{S32})$$

so the raw line density is $2b_{\text{eff}}W^2/[\pi(x^2 + W^2)]$ with total weight $2b_{\text{eff}}W$. Normalising to $\int \rho_1 dx = b_{\text{eff}}$ (dividing by the OSTZ diameter $2W$),

$$\boxed{\rho_1(x) = \frac{b_{\text{eff}}}{\pi} \frac{W}{x^2 + W^2}} \quad \left(\int \rho_1 dx = \frac{b_{\text{eff}}W}{\pi} \cdot \frac{\pi}{W} = b_{\text{eff}} \right), \quad (\text{S33})$$

the Lorentzian dislocation density. The same kernel is selected by the Peierls–Nabarro variational principle; here it follows from pure elastic mechanics, with no sinusoidal misfit energy as input.

S3.4 Step D: Fourier transform of the Lorentzian

With the convention $\hat{\rho}(k) = \int \rho e^{-ikx} dx$ and $k > 0$, close the contour in the lower half-plane (where e^{-ikx} decays). The denominator $x^2 + W^2 = (x-iW)(x+iW)$ has the single enclosed pole $x = -iW$, with residue $e^{-kW}/(-2iW)$; the clockwise contour gives $\oint = -2\pi i \sum \text{Res}$,

$$\int_{-\infty}^{\infty} \frac{e^{-ikx}}{x^2 + W^2} dx = -2\pi i \frac{e^{-kW}}{-2iW} = \frac{\pi}{W} e^{-kW}, \quad \hat{\rho}(k) = b e^{-|k|W}. \quad (\text{S34})$$

This pure exponential in k -space, evaluated at $k = 2\pi/b$, produces the $e^{-2\pi W/b}$ Peierls factor of Eq. (main): the OSTZ core width W filters the short-wavelength components of the lattice potential.

S3.5 Step E: cumulative slip and the Peierls–Nabarro profile

For N co-planar zones the densities superpose, $\rho_N = N\rho_1$, and the cumulative slip is

$$U(x) = \int_{-\infty}^x \rho_N(x') dx' = \frac{Nb_{\text{eff}}W}{\pi} \left[\frac{1}{W} \arctan \frac{x'}{W} \right]_{-\infty}^x = \frac{Nb_{\text{eff}}}{2} + \frac{Nb_{\text{eff}}}{\pi} \arctan \frac{x}{W}, \quad (\text{S35})$$

exactly the Peierls–Nabarro profile with $b = Nb_{\text{eff}} = N\gamma_0 W$ and core half-width $\zeta = W$ (Eq. main); $U(-\infty) = 0$, $U(+\infty) = b$.

S4 Mean-field decoupling of the ostz Hamiltonian

The OSTZ Hamiltonian [Eq. (main)] is $\mathcal{H} = \sum_i [\Delta F_0 - (\tau - \tau_0)\gamma_0 V_0] n_i - \frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j$. Writing $n_i = \bar{n} + \delta n_i$ and expanding, $n_i n_j = \bar{n}^2 + \bar{n} \delta n_i + \bar{n} \delta n_j + \delta n_i \delta n_j$, the Bragg–Williams approximation drops the fluctuation product $\delta n_i \delta n_j$, leaving

$$n_i n_j \approx n_i \bar{n} + n_j \bar{n} - \bar{n}^2. \quad (\text{S36})$$

Substituting into the interaction term, using $\sum_{j \neq i} J_{ij} = zJ_0$ (coordination z , mean coupling J_0), and absorbing the constant $\propto \bar{n}^2$,

$$-\frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j \approx -zJ_0 \bar{n} \sum_i n_i + \text{const}, \quad \mathcal{H}^{\text{MF}} = \sum_i h^{\text{eff}} n_i + \text{const}, \quad (\text{S37})$$

with $h^{\text{eff}} = \Delta F_0 - (\tau - \tau_0)\gamma_0 V_0 - zJ_0 \bar{n}$. The single-site partition function $\mathcal{Z}_i = 1 + e^{-h^{\text{eff}}/kT}$ gives the self-consistency relation

$$\bar{n} = \frac{e^{-h^{\text{eff}}/kT}}{1 + e^{-h^{\text{eff}}/kT}}, \quad h^{\text{eff}} = h^{\text{eff}}(\bar{n}), \quad (\text{S38})$$

which in the dilute, weak-coupling limit ($zJ_0 \bar{n} \ll kT$, $\bar{n} \ll 1$) reduces to $\bar{n} \approx e^{-\Delta F_0/kT} e^{(\tau - \tau_0)\gamma_0 V_0/kT}$.

S5 Partition-function route and detailed balance

The equilibrium occupation follows from $\langle N_{\text{active}} \rangle = kT \partial \ln \mathcal{Z} / \partial [(\tau - \tau_0)\gamma_0 V_0]$ with $\ln \mathcal{Z} = N_{\text{sites}} \ln(1 + e^{-h^{\text{eff}}/kT})$, giving $\langle N_{\text{active}} \rangle = N_{\text{sites}} \bar{n}$. This *static* route yields, in the dilute limit, a forward-bias-only factor $e^{2\Delta w/kT}$, whereas the kinetic Eyring balance of the main text gives $2 \sinh(\Delta w/kT)$: the two are complementary descriptions of the same process. The factor-of-two relation between the full work $2\Delta w = (\tau - \tau_0)\gamma_0 V_0$ in the Boltzmann energy difference and the half-work Δw in each kinetic barrier is required by detailed balance, $\Gamma^+/\Gamma^- = e^{2\Delta w/kT}$, consistent with the Hamiltonian. The hyperbolic-sine flow law [Eq. (main)] is an Eyring kinetic result and does not follow from the equilibrium partition function alone.

S6 Numerical-verification suite and the 41-system comparison

All algebraic identities and numerical values quoted in the main text are reproduced to the stated precision by an accompanying Jupyter notebook (`OSET_derivations.ipynb`), which executes automated cross-checks: the geometry identities $V_0 = \frac{4}{3}\pi\alpha W^3 = \frac{2}{3}\pi W^3$ and $V_0/A_0 = \frac{2}{3}W$; the Eshelby

values of Table S1; the Read–Shockley saturation identity $\gamma_{\text{GB}}(\theta_m) = E_0\theta_m$; the limit $\gamma_{\text{GB}} \rightarrow 0$ as $\theta \rightarrow 0$; and the per-system statistics below.

Table S2 gives the full 41-system comparison underlying Table 3 and Figs. 5–6 of the main text. Columns: γ_0, ε_0 (paper fit); ΔF_0 (OSET activation barrier, eV); Q (paper activation energy per event, eV); γ_B^{lit} (paper fit, J m^{-2}); γ_B^{OSET} (OSET Read–Shockley, J m^{-2}); ratio = $\gamma_B^{\text{lit}}/\gamma_B^{\text{OSET}}$.

Table S2: Full 41-system comparison: OSET (parameter-free) versus the semi-empirical constants of Harisankar and Padmanabhan [3].

#	System	γ_0	ε_0	ΔF_0	Q	γ_B^{lit}	γ_B^{OSET}	ratio
1	Zn–22Al (2.5 μm)	0.059	0.034	0.195	0.609	1.29	0.29	4.5
2	Zn–22Al (0.9 μm)	0.061	0.035	0.200	0.811	1.31	0.28	4.7
3	Al–33Cu–0.4Zr	0.066	0.038	0.263	1.449	1.61	0.31	5.2
4	Al–MgScMn (3 μm)	0.089	0.051	0.264	1.310	1.57	0.17	9.1
5	Al–MgScMn (1 μm)	0.083	0.048	0.260	1.274	0.92	0.19	4.8
6	Al–3Mg–0.2Sc	0.086	0.050	0.269	1.147	1.46	0.19	7.9
7	Al–8.9Zn–2.6Mg–Sc	0.080	0.046	0.267	1.299	1.47	0.21	6.8
8	Al–5Mg–Mn–Sc (24 μm)	0.090	0.052	0.266	1.370	0.87	0.17	5.2
9	Al–17Si–Fe–Mg–Cu–Ni	0.089	0.051	0.306	1.126	0.88	0.20	4.4
10	Mg–6Zn–0.8Zr	0.080	0.046	0.264	1.232	1.54	0.16	9.8
11	Mg–4Y–0.7Zr–Nd	0.087	0.050	0.263	1.265	1.35	0.13	10.2
12	Mg–5.8Zn–1Y–Zr	0.088	0.051	0.274	1.088	1.37	0.13	10.3
13	Ti–6Al–4V	0.116	0.067	0.585	2.491	1.14	0.20	5.6
14	Cu–2.8Al–Si–Co (7 μm)	0.096	0.056	0.394	1.499	1.48	0.27	5.5
15	Cu–2.8Al–Si–Co (3 μm)	0.095	0.055	0.392	1.504	0.99	0.28	3.6
16	IN836 (Cu–Ni–Zn)	0.081	0.047	0.290	1.272	1.34	0.27	5.0
17	Ti–43Al	0.144	0.088	0.505	2.939	1.45	0.10	14.2
18	Ti–48Al	0.138	0.080	0.461	2.962	1.20	0.11	11.3
19	Ti–46.2Al–2.2Cr	0.125	0.072	0.460	2.723	1.07	0.13	8.2
20	Co ₃ Ti	0.096	0.055	0.414	2.824	0.86	0.28	3.1
21	Ni ₃ Si	0.092	0.053	0.465	2.288	0.87	0.34	2.6
22	ZrO ₂	0.082	0.047	1.256	3.481	1.65	0.57	2.9
23	ZrO ₂ –3Y	0.088	0.051	1.261	3.790	0.85	0.49	1.7
24	ZrO ₂ –4Y	0.089	0.051	1.253	3.909	1.57	0.48	3.3
25	Al ₂ O ₃ –ZrO ₂ –mullite	0.067	0.039	1.130	3.473	0.86	0.75	1.1
26	Al ₂ O ₃ –NiAl ₂ O ₄ –ZrO ₂	0.064	0.037	1.244	3.690	1.35	0.91	1.5
27	6061 / 20% SiC	0.071	0.041	0.439	1.556	1.05	0.45	2.3
28	7075 / 20% SiC	0.069	0.040	0.421	1.537	0.97	0.45	2.1
29	Zr ₆₅ BMG	0.074	0.043	0.253	1.438	1.35	0.26	5.2
30	Zr _{52.5} BMG	0.075	0.043	0.268	1.518	0.78	0.27	2.9
31	La ₅₅ BMG	0.059	0.034	0.151	1.015	1.60	0.21	7.7
32	La ₆₀ BMG	0.073	0.042	0.213	0.963	0.82	0.19	4.3
33	Fe ₇₂ BMG	0.079	0.046	0.346	1.452	0.78	0.35	2.2
34	Limestone	0.073	0.042	1.045	2.554	0.57	0.29	2.0
35	Anorthite–Diopside (dry)	0.062	0.036	1.601	3.117	1.12	0.61	1.8
36	Anorthite–Diopside (wet)	0.062	0.036	1.624	2.697	1.27	0.62	2.1
37	Ice (10 μm)	0.115	0.067	0.169	0.332	0.79	0.03	31.0
38	Ice (1700 μm)	0.117	0.068	0.173	0.408	1.44	0.03	57.1

Table S2 (continued)

#	System	γ_0	ε_0	ΔF_0	Q	γ_B^{lit}	γ_B^{OSET}	ratio
39	Si ₃ N ₄	0.085	0.049	2.765	3.168	1.05	1.15	0.9
40	Zirconia–Alumina	0.069	0.040	1.311	3.166	0.94	0.83	1.1
41	Zirconia–Spinel	0.075	0.043	1.180	3.434	1.62	0.64	2.5

Summary statistics ($n = 41$). $\varepsilon_0 = 0.049 \pm 0.012$ (OSET: 0.05); $\gamma_0 = 0.085 \pm 0.020$ (OSET: 0.10–0.12); $Q/\Delta F_0 = 4.05 \pm 1.39$; γ_B ratio median 4.5 (15/41 within $3\times$, 24/41 within $5\times$, 35/41 within $10\times$); the ratio is uncorrelated with homologous temperature ($r = 0.00$). Ice (#37–38) is the only large outlier: its fitted γ_B exceeds the true ice grain-boundary energy ($\approx 0.065 \text{ J m}^{-2}$), so the OSET value is in fact closer to reality.

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